

# **JNTU Hyderabad**

## **MID-TERM EXAMINATION ? 2025**

CAD/CAM and Automation ? Mid-Term Examination, 2025

Subject Code: ME602 Semester: 6 Regulation: R20

Time: 90 Minutes Max. Marks: 30

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.  
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### **SECTION A (Unit 1: Geometric Modeling)**

1. Define Geometric Modeling. Explain its significance in CAD/CAM and Automation with a suitable example. [12 Marks]
2. How is Geometric Modeling implemented? Give an algorithm / procedure and explain. [12 Marks]
3. List the advantages and limitations of Geometric Modeling. Explain any two in detail. [8 Marks]

### **SECTION B (Unit 2: Finite Element Analysis Basics)**

4. Write short notes on Finite Element Analysis Basics. [12 Marks]
5. Solve a problem related to Finite Element Analysis Basics. Show all the steps clearly. [12 Marks]
6. Solve a problem related to Finite Element Analysis Basics. Show all the steps clearly. [12 Marks]

### **SECTION C (Unit 3: CNC Programming)**

7. Discuss the engineering applications of CNC Programming. [8 Marks]
8. Discuss the engineering applications of CNC Programming. [12 Marks]
9. Compare and contrast the key concepts involved in CNC Programming. [12 Marks]

### **SECTION D (Unit 4: Computer Aided Manufacturing)**

10. Write short notes on Computer Aided Manufacturing. [12 Marks]
11. Derive the expression related to Computer Aided Manufacturing and explain each term. [12 Marks]
12. Compare and contrast the key concepts involved in Computer Aided Manufacturing. [8 Marks]

### **SECTION E (Unit 5: Industrial Robotics)**

13. Compare and contrast the key concepts involved in Industrial Robotics. [12 Marks]
14. How is Industrial Robotics implemented? Give an algorithm / procedure and explain. [12 Marks]
15. Describe the working principle of Industrial Robotics with necessary equations. [12 Marks]

\*\*\* End of Question Paper \*\*\*